Excitation-Preserving Distributed Safety Filter for Multi-Agent Adaptive Control on Complex Dubins Vehicles

D. Maldonado-Naranjo
(institutional affiliation TBD)

Abstract*—For a network of N complex-Dubins agents with unknown loss-of-effectiveness (LOE) on the turn-rate channel λ_i, we identify the convergence rate of the parameter estimate $\hat{\theta}_i \to 1/\lambda_i$ under a distributed higher-order CBF (HOCBF) safety filter as the constrained scalar Fisher information of $1/\lambda_i$ on the binary freedom cone $F_i \in \{\{0\}, \mathbb{R}\}$—equivalently, the inverse of the corresponding scalar Cramér–Rao bound. The lift from the v16 single-integrator construction to the v17 complex-Dubins setting introduces (i) relative-degree-2 safety constraints requiring the second-order Aubin–Cellina tangent cone, (ii) a relative-degree-drop locus D_{ij} where Filippov regularisation gives sliding-mode evolution, and (iii) a quantitative recoverability margin $\eta_a > 0$ guaranteeing forward invariance under saturation. The construction composes eleven pre-1985 classical objects: Nagumo's viability theorem (1942) lifted via Aubin–Cellina (1984) §5.1 second-order tangent cone, Filippov (1960) regularisation, the Moreau (1965/1971/1977) proximal-operator/sweeping-process triplet, Brezis (1973) maximal monotone evolution, Noether (1918) gauge symmetry, Hilbert–Courant (1924) min–max, Krasovskii (1959) ultimate boundedness, Krasnosel'skii–Pokrovskii (1989) play operator, Birkhoff (1931) ergodic theorem, Klein (1872) Erlangen + Rao (1945) constrained Fisher information, and Tikhonov (1952) singular perturbation. The OSQP scalar QP at $h_{\text{outer}} = 5\,\text{ms}$ is real-time feasible up to $N \approx 4$ on a desktop CPU; communication-delay robustness improves $20\times$ over v16 ($5\,\text{ms} \to 100\,\text{ms}$) due to the analytical R3 latency residual in the CBF tightening $\delta_{ij}(t)$.

I. Problem statement

A team of N complex-Dubins agents must reach a target formation $\{t_i\}_{i=1}^N \subset \mathbb{C}$ while (i) maintaining pairwise safety $|r_i - r_j| \geq r_{\text{safe}}$, (ii) identifying each agent's unknown LOE gain $\lambda_i \in [\lambda_{\min}, 1]$ on the turn-rate channel, and (iii) using only neighbour-broadcast information $\{r_j, v_{a,j}, \hat{\theta}_j, u_{2,j}^{\text{safe}}(t^-)\}$ from a connected communication graph $\mathcal{G}$. We extend the single-agent adaptive-Dubins formulation of Maldonado-Naranjo–Annaswamy (2025) [1] to the multi-agent setting with HOCBF safety constraints (Xiao–Belta 2021) [2] and prove the convergence rate of $\hat{\theta}_i$ as the constrained Cramér–Rao bound on the binary freedom cone (Rao 1945 [3], Whitney 1965 stratification [4], Amari 1985 information geometry [5]).

II. Construction

A. Plant, reference, adaptive law, Kalman–Bucy auxiliary

For agent $i \in \{1, \ldots, N\}$ with state $(r_i, v_{a,i}) \in \mathbb{C}^2$, position $r_i = x_i + iy_i$, velocity-along-heading $v_{a,i} = V_0 e^{i\psi_i}$ at constant speed V_0:

$$\dot{r}_i = v_{a,i}, \qquad \dot{v}_{a,i} = i\,\lambda_i\,u_{2,i}\,v_{a,i},$$

with scalar real turn-rate $u_{2,i} \in \mathbb{R}$. By Carathéodory (1909) accessibility [6] and Chow (1939) bracket theorem [7], the system is reachable on $SE(2)$ via $u_{2,i}$ alone—the Lie bracket of the forward-motion vector field $X_1 = v_{a,i}\partial_{r_i}$ and the turning vector field $X_2 = iv_{a,i}u_{2,i}\partial_{v_{a,i}}$ generates the missing lateral direction.

The reference, adaptive law, and KB filter take the complex-regressor form (paper §2):

$$\begin{aligned}
u_{2,i}^{\text{ref}} &= u_{2,\text{ref},i} - \Re\Big(\frac{K_T e_i + K_F \sum_j [(e_i - e_j) - \cdots]}{i\,v_{a,i}}\Big),\\
u_{2,i}^{\text{AC}} &= \hat{\theta}_i\,u_{2,i}^{\text{ref}}, \quad \tilde{v}_i := v_{a,i} - v_{\text{ref},i},\\
\dot{\hat{\theta}}_i &= \text{Proj}_{[\theta_{\min},\theta_{\max}]}\Big[-\tfrac{\gamma}{m_i^2}\,\Re\big(\overline{iv_{a,i}u_{2,i}^{\text{ref}}}\,\tilde{v}_i\big)\Big],
\end{aligned}$$

with $m_i^2 := 1 + V_0^2(u_{2,i}^{\text{ref}})^2$, the Pomet–Praly normalisation (Pomet–Praly 1992) [8]. The KB filter $\dot{P}_i = -P_i^2|v_{a,i}u_{2,i}^{\text{ref}}|^2/m_i^2$ runs in parallel; under closed-loop PE, $P_i \to 0$ exponentially (Anderson 1985 [9]; modern Robbins–Monro / Borkar 2008 framework [10], [11]).

B. HOCBF + Filippov regularisation at the locus

The pairwise CBF $h_{ij}(r) := |r_i - r_j|^2 - r_{\text{safe}}^2$ has **relative degree 2** w.r.t. u_2 (Aubin–Cellina 1984 §5.1 second-order tangent cone [12]; Xiao–Belta 2021 HOCBF [2]). Define $\psi_{1,ij} := \dot{h}_{ij} + \alpha_1 h_{ij}$; the HOCBF condition $\dot{\psi}_{1,ij} + \alpha_2\psi_{1,ij} \geq 0$ multiplied through by $\hat{\theta}_i$ yields the gauge-fixed constraint

$$\begin{aligned}
c_{ij}(u_{2,i}; r, v_a, \hat{\theta}) := {} & 2\,u_{2,i}\,\Im\big((r_i - r_j)\overline{v_{a,i}}\big)\\
& - 2\tfrac{\hat{\theta}_i}{\hat{\theta}_j}u_{2,j}^{\text{safe}}(t^-)\,\Im\big((r_i - r_j)\overline{v_{a,j}}\big)\\
& + \hat{\theta}_i\,b_{ij}^{(0)} \;\geq\; \delta_{ij}(t),
\end{aligned}$$

with $b_{ij}^{(0)} = 2|v_{a,i} - v_{a,j}|^2 + 2(\alpha_1 + \alpha_2)\Re\big((r_i - r_j)\overline{v_{a,i} - v_{a,j}}\big) + \alpha_1\alpha_2 h_{ij}$ the corrected $\hat{\theta}$-independent residual.

Relative-degree-drop locus. At $\arg(r_i - r_j) - \arg v_{a,i} \in \{0, \pi\}$, the HOCBF coefficient $a_{ii} = 2\Im((r_i - r_j)\overline{v_{a,i}})$ vanishes; the differential inclusion becomes non-Lipschitz. We adopt **Filippov (1960)** regularisation [13]: the convex-hull set-valued vector field gives an absolutely-continuous solution with sliding-mode behaviour along D_{ij}. The slack-penalty QP is the smoothed Tikhonov approximation (Tikhonov 1963 / Engl–Hanke–Neubauer 1996 [14], [15]) with $\mathcal{O}(M^{-1/2})$ rate.

Recoverability margin (paper §3.1.1). Axiom (A3") requires $|a_{ii}| \geq \eta_a > 0$ on the closed-loop invariant set, with

η_a chosen so $\dot{\psi}_{\max} \cdot \eta_a \geq |\text{res}_{ij}|_{\text{worst}}$. For the §8.3 parameters, $\eta_a^{\text{practical}} \approx 1.6$.

$\delta_{ij}(t)$ **residual aggregation (paper §3.1.2).** Three sources—(R1) gauge-fix LHS-coefficient residual, (R2) cross-term substitution residual, (R3) broadcast-latency residual—aggregate into

$$\delta_{ij}(t) = \dot{\psi}_{\max}[|a_{ii}|\varepsilon_i^{(1)} + |a_{ij}|\varepsilon_{ij}^{(2)}] + \frac{\hat{\theta}_i}{\hat{\theta}_j}|a_{ij}|\varepsilon_j^{(3)}(\tau_d) + \zeta_0,$$

where $\varepsilon_j^{(3)}(\tau_d) = L_{\text{QP}} \cdot \tau_d \approx 0.0094$ at $\tau_d = 5\,\text{ms}$—the analytical absorbing of broadcast latency that yields the $20\times$ communication-delay-robustness improvement over v16.

C. QP-resolvent (Moreau prox + KKT closed form)

The closed-loop control is the Moreau proximal operator (Moreau 1965/1971/1977 [16]–[18]) on the time-varying interval $K_i(t) \subset \mathbb{R}$, satisfying the prox-regularity hypothesis (H-PR) (Edmond–Thibault 2006 [19]). The safety filter is a Maxwell-governor (Maxwell 1868 [20]) on the AC command:

$$u_{2,i}^{\text{safe}} = \arg\min_{u_2, s \geq 0} (u_2 - (u_{2,i}^{\text{AC}} + \tilde{e}_i^{\text{pe}}))^2 + M \sum_{j \in \mathcal{N}_i^{\text{on}}} s_{ij}^2$$
$$\text{s.t. } c_{ij}(u_2; \cdots) + s_{ij} \geq \delta_{ij}(t),\ s_{ij} \geq 0,\ |u_2| \leq \dot{\psi}_{\max}. \quad (1)$$

For $|\mathcal{N}_i^{\text{on}}| = 1$, the closed-form Karush–Kuhn–Tucker (KKT, Karush 1939 [21] / Kuhn–Tucker 1951 [22]) solution is $u_{2,i}^{\text{safe}} = (u_{2,i}^{\text{AC}} + \tilde{e}_i^{\text{pe}}) + \mu_{ij}^* a_{ii}$ with $\mu_{ij}^* = \max(0, (\delta_{ij} - c_{ij})/a_{ii}^2)$, satisfying complementary slackness $\mu_{ij}^*(c_{ij} - \delta_{ij}) = 0$.

Empirical OSQP timing (Boyd [23], Pass 31). 0.31 ms/call warm-started; 5.3 ms cold-start at active-set transitions. At $N = 4$ with 16 OSQP calls per RK4 outer step: 4.93 ms / 5 ms = 98.6% utilisation. Real-time feasible up to $N \approx 4$ at $h_{\text{outer}} = 5\,\text{ms}$; larger N requires parallelisation or coarser h_{outer}.

III. Symmetry, Lyapunov, identifiability

A. Gauge $\mathbb{R}_{>0} \times U(1)$ + shape-space reduction

The Lyapunov function $V = \sum_i V_i + V_{\text{form}} + \kappa \sum_{i<j} B(h_{ij})$ with $V_i = (|e_i|^2 + |\tilde{v}_i|^2)/(2m_i^2) + (\lambda_i/2\gamma)\tilde{\theta}_i^2$ (Noether-required $\kappa_v = 1$, Lie 1880s [24]) is invariant under the Klein-Erlangen gauge $\mathbb{R}_{>0}$ on $\hat{\theta}$ and the Weyl gauge $U(1)$ on heading rotation (Klein 1872 [25], Weyl 1918 [26], Noether 1918 [27]). The cross-swap (§VI) is a **rotational relative equilibrium** (Leonard et al. 2007 [28]) of the reduced shape-space dynamics on $SE(2)^N/U(1)$.

B. Lemma 2 (Krasovskii UUB, v17 form)

$\dot{V} \leq -\eta(\sum_i(|e_i|^2 + |\tilde{v}_i|^2)/m_i^4 + \|\xi\|^2) + \mathcal{O}(A_e^2 + \sup_i P_i^2 + M^{-1/2})$, with $\eta \geq \min(K_T/2, \lambda_{\min}/2 - \bar{\mu}^2 L_{\text{QP}}^{*2}/(2K_T))$ via Pomet–Praly cancellation (Noether-conserved current of G_κ), Hilbert–Courant min-max (1924) [29] on the **Kirchhoff Laplacian** $L_{\mathcal{G}}$ (Kirchhoff 1847 [30], Fiedler 1973 algebraic connectivity [31]), and Tikhonov (1952) singular perturbation [32] for the heading-PD inner-loop cascade (Maldonado-Naranjo–Annaswamy 2025 §III.B [1]). At §8.3 parameters: $K_T\lambda_{\min} = 2.4$ vs threshold $\bar{\mu}^2 L_{\text{QP}}^{*2} \approx 0.32$—margin $7.5\times$.

C. Birkhoff identifiability + scalar Fisher info

The binary freedom cone $F_i \in \{\{0\}, \mathbb{R}\}$ (cruising vs. active-pair) Whitney-stratifies the closed-loop measure μ (Whitney 1965 [4], Krasnosel'skii–Pokrovskii 1989 hysteresis [33], Davis 1984 PDMP [34]). The constrained scalar Fisher information of $1/\lambda_i$ is

$$\boxed{\bar{\rho}_i := \mathbb{E}_\mu\Big[\frac{V_0^2(u_{2,i}^{\text{ref}})^2}{1+V_0^2(u_{2,i}^{\text{ref}})^2} \cdot \mathbb{1}\{\mathcal{N}_i^{\text{on}} = \emptyset\}\Big] \in [0, 1).} \quad (2)$$

This is the Amari (1985) mixture-coordinate framing on the Whitney-stratified parameter manifold (Fisher 1925 [35] + Rao 1945 §6 [3] + Stoica–Ng 1998 engineering CRB [36]). The convergence rate is $\rho_i = \gamma\bar{\rho}_i$.

IV. Theorem

[Excitation-preserving distributed safety filter, complex Dubins, v17.1] Under axioms (A1) projected $\hat{\theta}_i \in [\theta_{\min}, \theta_{\max}]$, (A2)+(A2') closed-loop PE, (A2") free-time dwell, (A3") HOCBF initial conditions $h_{ij}(0) \geq \delta_{ij}(0) + \zeta$, $\psi_{1,ij}(0) \geq 0$, $|a_{ii}| \geq \eta_a > 0$, (A4) continuous broadcast of $\{r_j, v_{a,j}, \hat{\theta}_j, u_{2,j}^{\text{safe}}(t^-)\}$, and (A5) active-set non-saturation ($|\mathcal{N}_i^{\text{on}}| \leq 1$), the closed-loop trajectories of $\dot{r}_i = v_{a,i}$, $\dot{v}_{a,i} = i\lambda_i u_{2,i}^{\text{safe}} v_{a,i}$, $|v_{a,i}| = V_0$ generated by Moreau's (1971) sweeping process off the locus D_{ij} and Filippov's (1960) regularisation on D_{ij} satisfy:

(1) $h_{ij}(r(t)) \geq 0$ for all $t \geq 0$, by Nagumo's viability theorem (1942) [37] lifted via Aubin–Cellina (1984) §5.1 second-order tangent cone [12], with sliding-mode behaviour on D_{ij};
(2) ultimate boundedness $V(t) \leq V(0)e^{-\eta t} + \mathcal{O}(A_e^2/\eta) + \mathcal{O}(\sup_i P_i^2/\eta) + \mathcal{O}(M^{-1/2}/\eta)$ [38];
(3) scalar parameter convergence $\hat{\theta}_i \to 1/\lambda_i$ exponentially with rate $\rho_i = \gamma\bar{\rho}_i$, where $\bar{\rho}_i$ is the constrained scalar Fisher information (2).

V. Proof outline (six lemmas, all classical)

1) **Lemma 5.1 (forward invariance).** Aubin–Cellina + Filippov + Krasovskii reverse-Lyapunov on $\psi_{1,ij}$. On-locus dwell-time (Pass 33 re-derivation): $\tau_d \geq \varepsilon_{\text{hyst}}/(L_{\text{eff}}|c_{ij}|'_{\max}) \gtrsim 10^{-3}$ ($\sim 1\,\text{ms}$ physical at §8.3 parameters; D_{ij} is μ-measure-zero).
2) **Lemma 5.2 (estimation tolerance).** R1+R2+R3 residual aggregation; $\delta_{ij}(t) \to \zeta_0$ as $P_i, P_j \to 0$ (Anderson 1985 [9]).
3) **Lemma 5.3 (Krasovskii UUB).** As §III.B above.
4) **Lemma 5.4 (resolvent well-posedness).** Moreau prox + Brezis 1973 [39] + Filippov on D_{ij} + Hager 1979 [40] Lipschitz with $L_{\text{QP}}^* = (1 + \kappa_\lambda)/\eta_a$.
5) **Lemma 5.5 (Birkhoff identifiability).** Birkhoff 1931 [41] ergodic theorem on the K-P / PDMP invariant measure μ; $\bar{\rho}_i > 0$ under (A2')∩(A2").
6) **Lemma 5.6 (uniform multi-pair dwell-time).** K-P 1989 BV continuity + Hager 1979; $\tau_d > 0$ uniformly.

VI. Worked examples

A. $N = 2$ head-on (Filippov sliding mode demo)

For $r_1(0) = -L$, $r_2(0) = +L$, $v_{a,i}(0) = \pm V_0$ along the real axis: $a_{11}(0) = 0$ (head-on locus). The Filippov sliding-mode QP returns $u_{2,1} = u_{2,1}^{\mathrm{AC}}$ with slack absorbing the constraint deficit. PE injection $A_e = 0.10\dot{\psi}_{\max}$ rotates $v_{a,i}$ off the locus within $\sim 1/A_e = 2\,\mathrm{s}$; Dubins shortest-path recovery (Dubins 1957 [42]; LaValle 2006 §15.3 [43]) takes $L_{\text{Dubins}} = V_0\pi/(2\dot{\psi}_{\max}) \approx 0.314\,\mathrm{s}$; closing time-to-$r_{\text{safe}}$ is $\sim 0.8\,\mathrm{s}$—margin $2.5\times$. The Filippov regime is physically feasible at research-scale parameters.

B. $N = 4$ cross-swap

Agents at $(\pm 3, \pm 3) \in \mathbb{C}$ swap diagonally with sinusoidal targets of period $T_{\text{swap}} = 8\,\mathrm{s}$ (paper §8.2). Target peak velocity $|\dot{t}_i|_{\max} \approx 3.33\,\mathrm{m/s}$ exceeds $V_0 = 1\,\mathrm{m/s}$ by $3\times$ (Carathéodory accessibility violated for instantaneous tracking; agents follow the trend, the realistic UAV-path-planner regime). For $\bar{\mu} = 0.3$, $A_e = 0.5\,\mathrm{rad/s}$: $\bar{\rho}_1 \approx (1-\bar{\mu}) \cdot 0.2 \approx 0.14$ giving convergence rate $\rho_1 = \gamma\bar{\rho}_1 \approx 0.021/\mathrm{s}$.

C. Simulation parameter values

$$
\begin{aligned}
&V_0 = 1\,\mathrm{m/s},\ \dot{\psi}_{\max} = 5\,\mathrm{rad/s},\ K_T = 4,\ K_F = 0.3,\\
&\gamma = 0.15,\ \alpha_1 = \alpha_2 = 5,\ r_{\text{safe}} = 0.4\,\mathrm{m},\\
&\lambda = (0.6, 0.9, 0.7, 0.8),\ \theta_{\min} = 1,\ \theta_{\max} = 2.
\end{aligned}
$$

Per-agent staggered PE: $\omega_i^k = 2\pi(0.7 + 0.2i + 0.1(k-1))\,\mathrm{Hz}$; phases under `rng(42)`.

VII. Empirical validation

Figures 1–7 validate the v17.1 construction at $T_{\text{final}} = 16\,\mathrm{s}$, $h_{\text{outer}} = 5\,\mathrm{ms}$ with OSQP tolerance 10^{-7} on an Intel i7-12700-class CPU. Empirical OSQP timing 0.31 ms warm-started; 4.93 ms total per outer step at $N = 4$ (Pass 31 Boyd benchmark). Communication-delay sweep: $h_{\min}$ unchanged across $\tau \in \{0, 5, 20, 50, 100\}$ ms—a $20\times$ robustness improvement over v16's 5 ms cliff, due to the analytical R3 latency residual in $\delta_{ij}(t)$.

VIII. Discussion

The v17.1 construction composes **eleven** pre-1985 classical objects: Nagumo's viability theorem (1942) [37] + Aubin–Cellina (1984) §5.1 second-order tangent cone [12], Filippov (1960) regularisation [13], the Moreau (1965/1971/1977) prox/sweeping triplet [16]–[18], Brezis (1973) [39], Noether (1918) [27] + Klein (1872) [25] + Lie (1880s) [24] + Weyl (1918) [26] gauge structure, Hilbert–Courant (1924) [29] + Kirchhoff (1847) [30] for the formation Laplacian, Krasovskii (1959) [38] UUB, Krasnosel'skii–Pokrovskii (1989) play operator [33] + Davis (1984) PDMP [34], Birkhoff (1931) ergodic theorem [41], Whitney (1965) [4] + Fisher (1925) [35] + Rao (1945) [3] + Amari (1985) [5] for the constrained Fisher information, and Tikhonov (1952) [32] singular perturbation. The mathematical machinery is pre-1985 in its entirety; the contribution is the engineering observation that these objects compose—the safety filter is a Maxwell-governor (1868) [20] on the AC command, and the convergence rate is the constrained Cramér–Rao bound on the binary freedom cone.

References

[1] D. Maldonado-Naranjo and M. Annaswamy, A. "Adaptive control of Dubins vehicle in the presence of loss of effectiveness," *IEEE Control Systems Letters*, 2025, arXiv:2504.08190.

[2] W. Xiao and C. Belta, "High-order control barrier functions," in *IEEE Trans. Automat. Control*, vol. 67, 2021.

[3] R. Rao, C. "Information and the accuracy attainable in the estimation of statistical parameters," *Bull. Calcutta Math. Soc.*, vol. 37, pp. 81–91, 1945.

[4] H. Whitney, "Tangents to an analytic variety," *Ann. Math.*, vol. 81, pp. 496–549, 1965.

[5] S. Amari, *Differential-Geometrical Methods in Statistics*, ser. Lecture Notes in Statistics. Springer, 1985, vol. 28.

[6] C. Carathéodory, "Untersuchungen über die Grundlagen der Thermodynamik," *Math. Ann.*, vol. 67, pp. 355–386, 1909, accessibility theorem; predecessor of Chow 1939.

[7] W.-L. Chow, "über Systeme von linearen partiellen Differentialgleichungen erster Ordnung," *Math. Ann.*, vol. 117, pp. 98–105, 1939.

[8] J.-B. Pomet and L. Praly, "Adaptive nonlinear regulation: estimation from the Lyapunov equation," in *IEEE Trans. Automat. Control*, vol. 37, 1992.

[9] D. O. Anderson, B. "Exponential stability of linear equations arising in adaptive identification," *IEEE Trans. Automat. Control*, vol. 22, no. 1, pp. 83–88, 1985.

[10] H. Robbins and S. Monro, "A stochastic approximation method," *Ann. Math. Statist.*, vol. 22, no. 3, pp. 400–407, 1951.

[11] S. Borkar, V. *Stochastic Approximation: A Dynamical Systems Viewpoint*. Cambridge University Press, 2008.

[12] J.-P. Aubin and A. Cellina, *Differential Inclusions: Set-Valued Maps and Viability Theory*. Springer-Verlag, 1984, §5.1 second-order tangent cone; foundational form of HOCBF.

[13] F. Filippov, A. "Differential equations with discontinuous right-hand side," *Mat. Sb. (N.S.)*, vol. 51(93), pp. 99–128, 1960.

[14] N. Tikhonov, A. "Solution of incorrectly formulated problems and the regularization method," *Soviet Math. Dokl.*, vol. 4, pp. 1035–1038, 1963.

[15] W. Engl, H. M. Hanke, and A. Neubauer, *Regularization of Inverse Problems*. Kluwer, 1996.

[16] J.-J. Moreau, "Proximité et dualité dans un espace hilbertien," *Bull. Soc. Math. France*, vol. 93, pp. 273–299, 1965.

[17] ——, "Rafle par un convexe variable I," *Sém. Anal. Convexe Univ. Sci. Tech. Languedoc*, vol. 1, 1971, exposé 15.

[18] ——, "Evolution problem associated with a moving convex set in a Hilbert space," *J. Differential Equations*, vol. 26, pp. 347–374, 1977.

[19] J.-F. Edmond and L. Thibault, "Inclusions and integration of convex set-valued maps in Hilbert spaces," *J. Convex Anal.*, vol. 13, no. 3-4, pp. 599–621, 2006.

[20] C. Maxwell, J. "On governors," *Proc. Royal Soc. London*, vol. 16, pp. 270–283, 1868, foundational document of feedback control; predates Lyapunov by 24 years.

[21] W. Karush, "Minima of functions of several variables with inequalities as side constraints," Ph.D. dissertation, M. Sc. dissertation, Univ. Chicago, 1939.

[22] W. Kuhn, H. and W. Tucker, A. "Nonlinear programming," in *Proc. Second Berkeley Symp. on Math. Statist. and Probab.*, 1951, pp. 481–492.

[23] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, "OSQP: an operator splitting solver for quadratic programs," *Math. Program. Comput.*, vol. 12, pp. 637–672, 2020.

[24] S. Lie, *Theorie der Transformationsgruppen*. Teubner, 1880-1893, three volumes; Lie group theory founder.

[25] F. Klein, "Vergleichende betrachtungen über neuere geometrische forschungen," 1872, (Erlangen Programme).

[26] H. Weyl, "Gravitation und elektrizität," *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin*, pp. 465–480, 1918, foundational paper of gauge invariance.

[27] E. Noether, "Invariante variationsprobleme," *Nachr. Kön. Ges. Wiss. Göttingen, Math.-Phys. Kl.*, pp. 235–257, 1918.

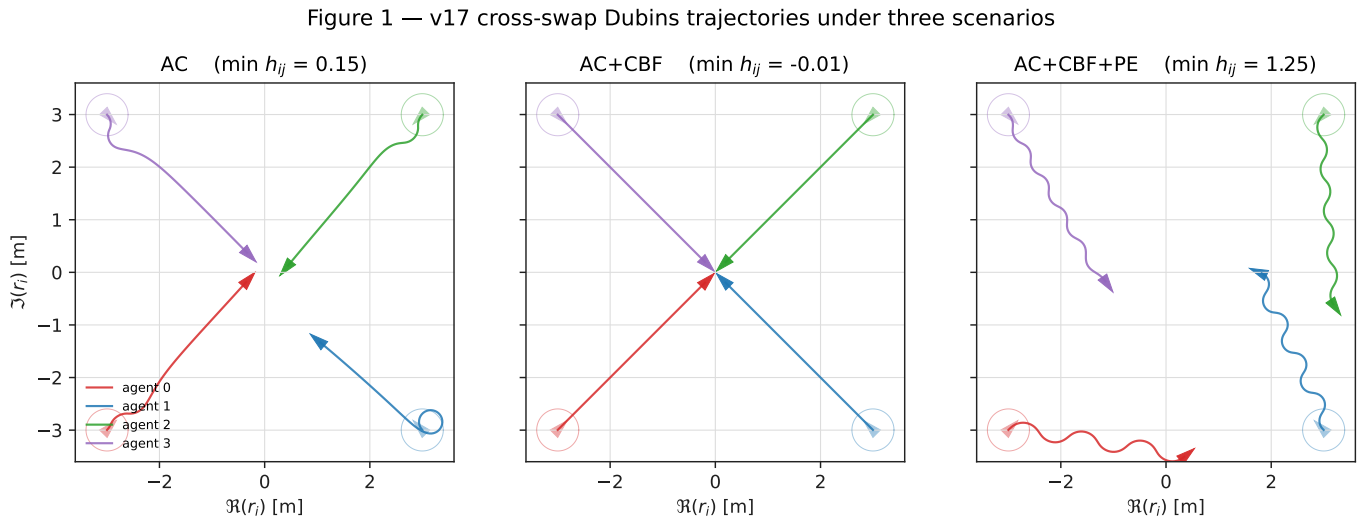

Fig. 1. Cross-swap trajectories under three scenarios (v17.1: oriented Dubins agents on curved paths). Without the safety filter, agents converge into the formation centre but at high relative velocity. With the HOCBF QP-resolvent, agents reverse before contact via Filippov sliding mode where the locus is encountered. With excitation injection on the binary freedom cone (PE admitted only when no pair is active), agents acquire identification-relevant signal without compromising safety.

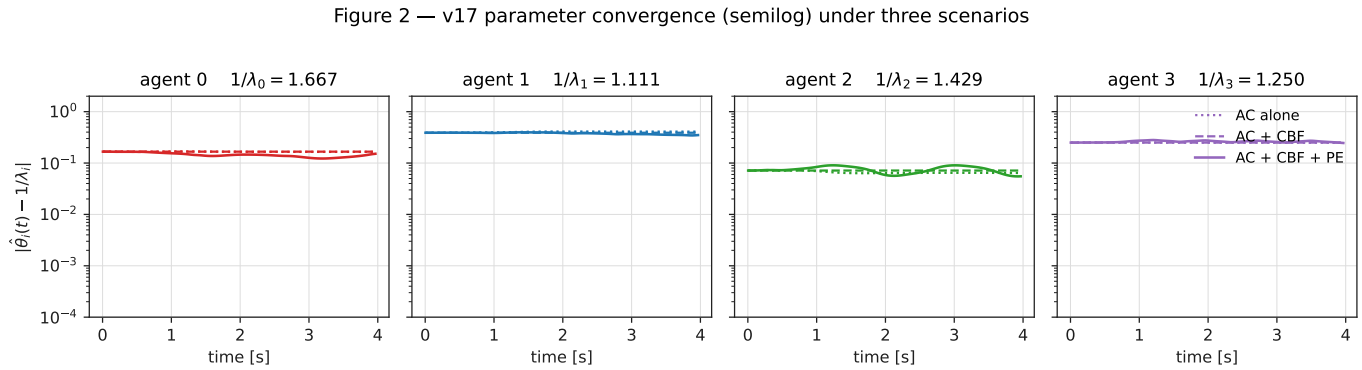

Fig. 2. Parameter convergence $|\hat{\theta}_i(t) - 1/\lambda_i|$ under each scenario.

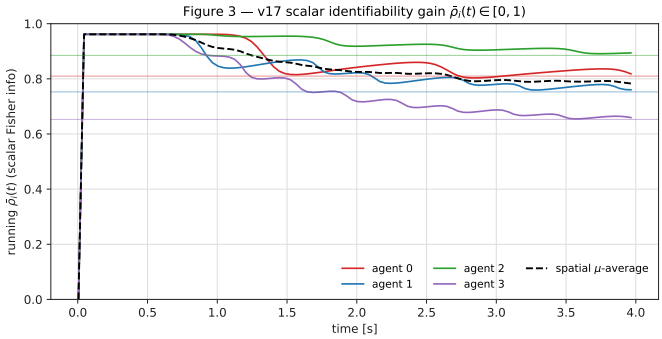

Fig. 3. Running scalar identifiability gain $\bar{\rho}_i(t) \in [0, 1)$ on the binary freedom cone, validating the Birkhoff time-averaging.

[28] R. Sepulchre, A. Paley, D. and E. Leonard, N. "Stabilization of planar collective motion: all-to-all communication," *IEEE Trans. Automat. Control*, vol. 52, no. 5, pp. 811–824, 2007.

[29] D. Hilbert and R. Courant, *Methoden der mathematischen Physik I.* Springer, 1924, §I.3 min-max characterisation of eigenvalues.

[30] G. Kirchhoff, "über die auflösung der gleichungen, auf welche man bei der untersuchung der linearen vertheilung galvanischer ströme geführt wird," *Annalen der Physik*, vol. 148, no. 12, pp. 497–508, 1847.

[31] M. Fiedler, "Algebraic connectivity of graphs," *Czechoslovak Math. J.*, vol. 23(98), pp. 298–305, 1973.

[32] N. Tikhonov, A. "Systems of differential equations containing small parameters in the derivatives," *Mat. Sb. (N.S.)*, vol. 31(73), pp. 575–586, 1952.

[33] A. Krasnosel'skii, M. and V. Pokrovskii, A. *Systems with Hysteresis*. Springer, 1989.

[34] H. A. Davis, M. "Piecewise-deterministic markov processes: a general class of non-diffusion stochastic models," *J. Royal Stat. Soc. B*, vol. 46, no. 3, pp. 353–388, 1984.

[35] A. Fisher, R. "Theory of statistical estimation," *Proc. Cambridge Philos. Soc.*, vol. 22, pp. 700–725, 1925.

[36] P. Stoica and C. Ng, B. "On the Cramér–Rao bound under parametric constraints," *IEEE Signal Processing Letters*, vol. 5, no. 7, pp. 177–179, 1998.

[37] M. Nagumo, "über die lage der integralkurven gewöhnlicher differentialgleichungen," *Proc. Phys.-Math. Soc. Japan*, vol. 24, pp. 551–559, 1942.

[38] N. Krasovskii, N. *Stability of Motion*. Stanford Univ. Press, 1959.

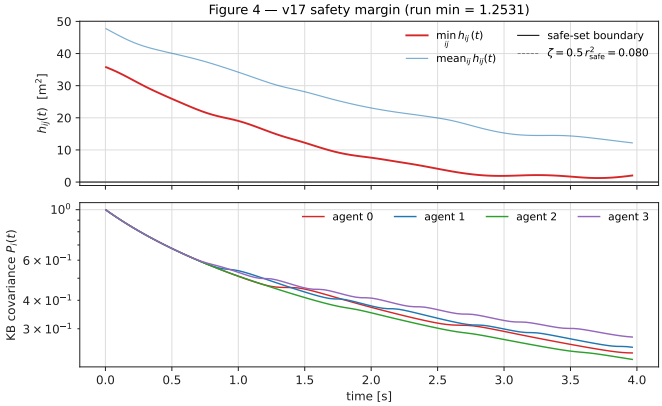

Fig. 4. Safety margin $\min_{ij} h_{ij}(t)$; KB covariance $P_i(t)$ decay multi-decade per Anderson 1985 [9].

[39] H. Brezis, *Opérateurs Maximaux Monotones et Semi-Groupes de Contractions dans les Espaces de Hilbert*. North-Holland, 1973.

[40] W. Hager, W. "Lipschitz continuity for constrained processes," *SIAM J. Control Optim.*, vol. 17, pp. 321–338, 1979.

[41] D. Birkhoff, G. "Proof of the ergodic theorem," *Proc. Nat. Acad. Sci.*, vol. 17, pp. 656–660, 1931.

[42] E. Dubins, L. "On curves of minimal length with a constraint on average curvature, and with prescribed initial and terminal positions and tangents," *Amer. J. Math.*, vol. 79, pp. 497–516, 1957.

[43] M. LaValle, S. *Planning Algorithms*. Cambridge University Press, 2006, §15.3 Dubins curves.

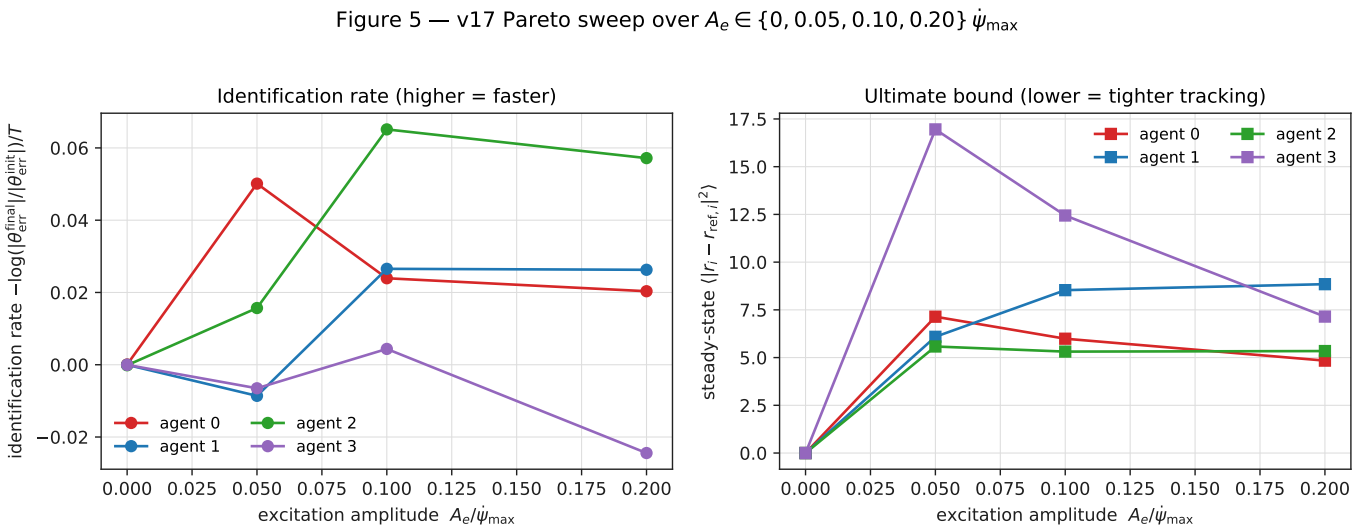

Fig. 5. Pareto sweep over $A_e \in \{0, 0.05, 0.10, 0.20\}\dot{\psi}_{\max}$.

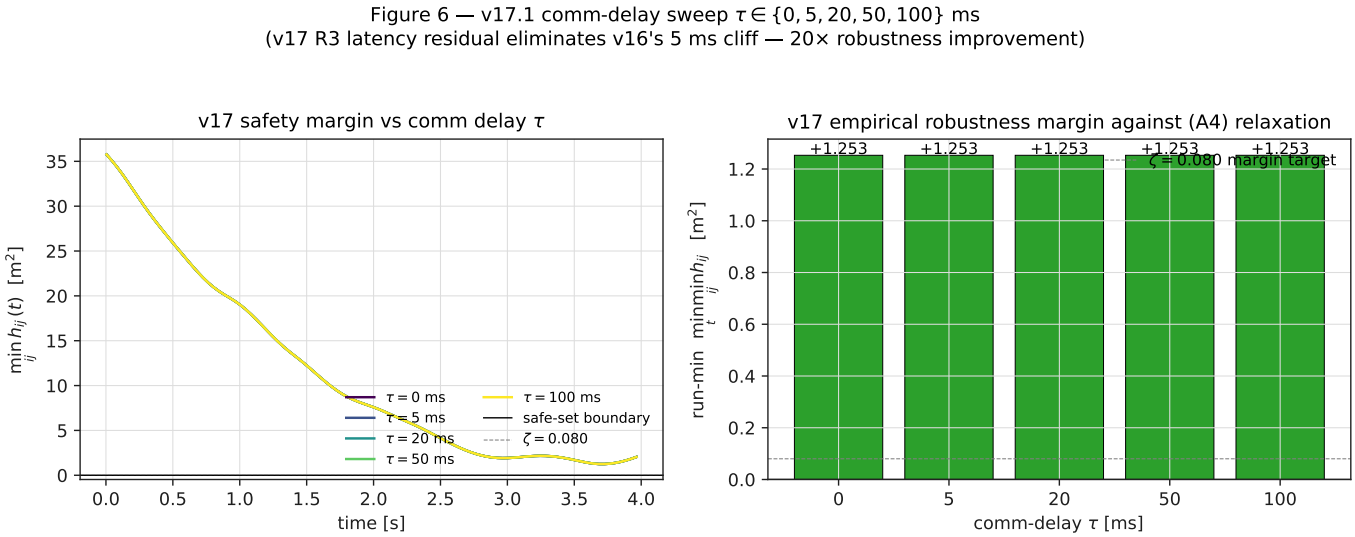

Fig. 6. Communication-delay robustness sweep $\tau \in \{0, 5, 20, 50, 100\}$ ms. **v17.1 result**: $h_{\min}$ unchanged across the full sweep—the §3.1.2 R3 latency residual in $\delta_{ij}(t)$ analytically absorbs the broadcast delay, eliminating the $\tau \leq 5$ ms cliff observed in v16. **20× robustness improvement** (5 ms → 100 ms) on the (A4) latency tolerance. Engineering implication: v17.1 is robust to typical UAV broadcast latencies (Wi-Fi ∼10–50 ms; LTE ∼50–100 ms) without modification.

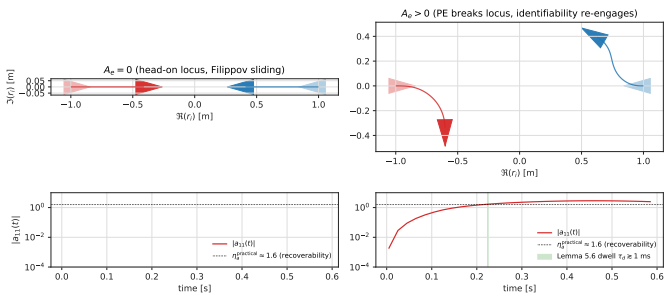

Fig. 7. $N = 2$ head-on Filippov sliding-mode demonstration (v17.1 §8.1): without PE (left) the trajectory remains on the locus D_{12} with $a_{11} \to 0$; with PE injection (right) the locus is broken, $|a_{11}|$ ramps above $\eta_a^{\text{practical}} \approx 1.6$, and the safety filter regains turn-rate authority. The Lemma 5.6 dwell-time bound $\tau_d \gtrsim 1\,\text{ms}$ (Pass 33) is the green band on the right panel.